

The Temperature Semiring

$(R_T, \oplus_T, \otimes, 0, 1)$

A fully hand-worked numerical example — one running case carried through every T regime.

Name / date: _____ Margin ruled for annotation →

1 The two operations, in PIC terms

Two roles for the two operations — this is the whole grammar:

- $\otimes = + -$ **within a coalition**. A coalition S collects incidences (log-scores) along it; its value is their sum.
- $\oplus_T -$ **across coalitions**. The aggregate / decode over all coalitions. Its temperature $T \geq 0$ is the only knob.

Take three coalitions with explicit incidence weights:

Table 1 — coalition values computed with $\otimes = +$

Coalition	incidences	value $v = \otimes$ (sum)
A	$\{2.0, 0.5\}$	$2.0 + 0.5 = 2.5$
B	$\{1.0, 1.0\}$	$1.0 + 1.0 = 2.0$
C	$\{3.0, -\infty\}$	$3.0 + (-\infty) = -\infty$

C shows the **annihilator**: one forbidden incidence $(-\infty, \text{probability } 0)$ kills the whole product. Then $-\infty$ switches hats and acts as the **additive identity**:

$$x \oplus_T (-\infty) = T \log(e^{x/T} + e^{-\infty}) = T \log e^{x/T} = x,$$

so C silently drops out of the aggregate. The same symbol $-\infty$ is annihilator for $\otimes$ and identity for $\oplus_T$ — exactly the “0 annihilates” axiom. The calculation therefore reduces to

$$Z_T = 2.5 \oplus_T 2.0 \oplus_T (-\infty) = 2.5 \oplus_T 2.0,$$

with margin $\delta = 2.5 - 2.0 = 0.5$ between the two surviving coalitions.

2 One identity drives every regime

Factor out the larger value ($a = 2.5$):

$$a \oplus_T b = T \log(e^{a/T} + e^{b/T}) = \underbrace{\max(a, b)}_{\text{tropical part}} + \underbrace{T \log(1 + e^{-\delta/T})}_{\text{excess over the max}}.$$

The entire story lives in that **excess term**: it is 0 at $T \rightarrow 0$, and grows to $T \log 2$ as $T \rightarrow \infty$.

3 Sweeping T — actual arithmetic

Using

$$e^{1.0} = 2.718, e^{1.25} = 3.490, e^{2.0} = 7.389, e^{2.5} = 12.182, e^{4.0} = 54.598, e^{5.0} = 148.413$$

$T \rightarrow 0^+$ (tropical / max-plus):

$$2.5 \oplus_0 2.0 = \max(2.5, 2.0) = 2.5, \quad \text{excess} = 0.$$

$T = 0.5$: scale up by $1/T = 2$ first.

$$0.5 \log(e^{5.0} + e^{4.0}) = 0.5 \log(203.011) = 0.5(5.3132) = 2.6566.$$

Check excess: $0.5 \log(1 + e^{-1.0}) = 0.5 \log(1.3679) = 0.1566. \checkmark$

$T = 1$ (log-semiring):

$$\log(e^{2.5} + e^{2.0}) = \log(19.572) = 2.9741.$$

Check excess: $\log(1 + e^{-0.5}) = \log(1.6065) = 0.4741. \checkmark$

$T = 2$:

$$2 \log(e^{1.25} + e^{1.0}) = 2 \log(6.2086) = 2(1.8259) = 3.6519.$$

Check excess: $2 \log(1 + e^{-0.25}) = 2 \log(1.7788) = 1.1519. \checkmark$

$T \rightarrow \infty$: excess $\rightarrow T \log 2$ diverges; the *value* stops being the interesting object — the *distribution* does (§5).

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Table 2 — the aggregate $2.5 \oplus_T 2.0$ across temperatures

T	$2.5 \oplus_T 2.0$	excess over max	reading
0^+	2.5000	0	argmax — A wins; margins, power diagrams
0.5	2.6566	0.1566	sharp softmax, near-geometric
1	2.9741	0.4741	log-partition over coalitions (probabilistic)
2	3.6519	1.1519	warm decode, coalitions blur
∞	$\rightarrow \max + T \log 2$	$\rightarrow T \log 2$	uniform mixing, max entropy

4 What the regimes *mean*: the induced decode

The probabilistically interesting quantity is the decode distribution $p_i \propto e^{v_i/T}$ — the softmax whose normalizer is Z_T . For example at $T = 1$:

$$p_A = \frac{e^{2.5}}{e^{2.5} + e^{2.0}} = \frac{12.182}{19.572} = 0.622.$$

Table 3 — decode distribution $p_i \propto e^{v_i/T}$; entropy H (nats) and perplexity PP rise monotonically with T

T	p_A	p_B	H	PP	shape
0^+	1.000	0.000	0.000	1.00	point mass (argmax)
0.5	0.731	0.269	0.582	1.79	sharp
1	0.622	0.378	0.663	1.94	native softmax
2	0.562	0.438	0.685	1.98	high-entropy (<i>not</i> uniform)
∞	0.500	0.500	0.693	2.00	uniform = max H

Definitions. H is the Shannon entropy $H(p_T) = - \sum_v p_T(v) \log p_T(v)$ (in nats); the perplexity $PP = e^H$ is the *effective number of live tokens*. H rises *monotonically* with T , from 0 (a point mass on the argmax) up to its ceiling $\log 2 \approx 0.693$ nats ($PP = 2$), reached only in the limit $T \rightarrow \infty$ where the mass is uniform over the two surviving coalitions $\{A, B\}$. So $T = 2$ ($H = 0.685$, $PP = 1.98$) is high-entropy and *close to* uniform — but not equal to it; “flattened” means *moved up the entropy curve*, not “made

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uniform.” And because C was annihilated to $-\infty$ it stays at probability 0 for every T : temperature flattens only within the live support, so the ceiling is $\log 2$, not $\log |V|$ over the full vocabulary.

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Lower T pulls the mass onto the winning coalition (the proofs’ world — pick the margin); higher T spreads it (the model in operation — temperature-controlled decoding). Same object, deformed: **Maslov dequantization**.

5 Idempotency — the algebraic fingerprint of T

Combine a coalition with an identical copy ($\delta = 0$), value $a = 2.5$:

$$a \oplus_T a = T \log(2e^{a/T}) = a + T \log 2.$$

- $T \rightarrow 0$: $2.5 \oplus_0 2.5 = 2.5$. **Idempotent** — geometry ignores multiplicity; a duplicate face is still one face.
- $T = 1$: $2.5 \oplus_1 2.5 = 2.5 + \log 2 = 3.1931$. **Not idempotent** — probability *counts* the copy; two equally-likely incidences double the mass, and $\log(2\times) = +\log 2$.

That $+\log 2$ is literally “the algebraic difference between geometry and probability.”

6 Distributivity — the load-bearing law, checked

$a \otimes (b \oplus_T c) = (a \otimes b) \oplus_T (a \otimes c)$. Take common factor $a = 2.0$, branches $b = 2.5$, $c = 2.0$, at $T = 1$:

LHS: $2.0 \otimes (2.5 \oplus_1 2.0) = 2.0 + 2.9741 = 4.9741$.

RHS:

$$(2.0+2.5) \oplus_1 (2.0+2.0) = 4.5 \oplus_1 4.0 = \log(e^{4.5} + e^{4.0}) = \log(144.615) = 4.9741.$$

✓ exact match with the LHS.

Pulling a shared log-weight out of the sum is exact at every T . At $T = 0$ it is the tropical statement $\max(a+b, a+c) = a + \max(b, c)$:

$$\max(4.5, 4.0) = 4.5 = 2.0 + \max(2.5, 2.0). \checkmark$$

Why this is the ReLU $\leftrightarrow$ tropical-polynomial rewrite. A ReLU is $\max(\cdot, 0)$; a stack multiplies (adds, in $\otimes$) such gates. Distributivity

expands a *product-of-maxes* into a *max-of-sums*:

$$(\max(p, q)) \otimes (\max(r, s)) = \max(p, q) + \max(r, s) = \max(p+r, p+s, q+r, q+s).$$

Numbers $p = 2, q = 0, r = 1, s = 0$: LHS = $\max(2, 0) + \max(1, 0) = 3$;

RHS = $\max(3, 2, 1, 0) = 3$. ✓ The four monomials

$\{p+r, p+s, q+r, q+s\}$ are the four coalitions — each picks one branch per factor, $\otimes$ adds incidences inside it, $\oplus_T$ aggregates across them.

ONE-LINE TAKEAWAY

Every line above is the single identity

$a \oplus_T b = \max(a, b) + T \log(1 + e^{-\delta/T})$ read at different T : excess $0 \rightarrow$ pure geometry (argmax, idempotent, power diagrams); excess $\log(1+e^{-\delta})$ at $T=1 \rightarrow$ probability (softmax, log-partition, $+\log 2$ on duplicates); excess $\rightarrow T \log 2 \rightarrow$ uniform decode. PIC fixes $\otimes = +$ and dequantizes the sum.

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Temperature in practice — from the semiring to the API

The single knob T of §1–§6 is exactly the *sampling temperature* of a real LLM API. Here is how it maps to practice — sensible values, when to exceed 1, and what $T \rightarrow \infty$ actually does.

A The scale is a provider convention, not a law

The decode distribution of §4 is literally the softmax over logits with temperature T :

$$p_T(v) = \frac{e^{L(v)/T}}{\sum_w e^{L(w)/T}}, \quad T \in [0, \infty).$$

$T = 0$ is greedy / argmax; $T = 1$ is the model’s native, as-trained distribution; $T \rightarrow \infty$ is uniform. There is no built-in ceiling — each provider just picks a cap and a default and exposes that number. The “0 to 1” intuition is in fact **Anthropic’s** convention; OpenAI runs 0 to 2.

Table A1 — temperature ranges differ by provider (caps change; verify in current docs)

Provider	Range	Default	Note
Anthropic (Claude)	0 – 1.0	1.0	Cannot flatten past the native distribution via the API.
OpenAI (GPT-4o etc.)	0 – 2.0	1.0	Values > 1 are allowed and meaningful.
Google Gemini (1.5 / 2.0)	0 – 2.0	~1.0	Older PaLM / Gemini 1.0 was 0 – 1.
Open weights (llama.cpp, vLLM, Ollama)	0 – ∞	varies	You set the raw divisor; crank it arbitrarily high.

Two consequences worth holding onto:

- **The same number need not mean the same sharpness across providers.** Internal normalisation is not standardised, so **0.7** on one model $\neq$ **0.7** on another. Calibrate per model.
- **On Claude, $T = 1$ is as random as it goes** (the native distribution); OpenAI and open models give headroom above native.

B Sensible operating bands

A working band of **0.3–0.9** is fine, and even conservative. For reliability-critical work — extraction, classification, structured / JSON output, coding, tool-calls — most people go *lower* (**0–0.2**). And when A/B-testing prompt versions, hold **T** low and fixed so sampling noise does not swamp the signal you are measuring.

Table A2 — rough task $\rightarrow$ temperature guide

Task	Temperature
Extraction, classification, structured / JSON, tool-calls	0 – 0.2
Factual QA, RAG, coding	0.2 – 0.5
General assistant / drafting	0.5 – 0.8
Creative, varied, “give me options”	0.8 – 1.0
Max-diversity candidates, then filter	1.0 – 1.5 *

* Only on providers that allow $T > 1$ — not the Anthropic API.

C When to reach above $T = 1$

By Proposition 4.1, raising **T** never changes which token wins the argmax — it only reshapes the field behind the winner. So **$T > 1$** earns its keep precisely where you *consume the distribution* rather than the top token, and have something downstream to catch misfires:

- **Best-of- n / sample-and-rerank** — many diverse candidates, then score and keep the best.
- **Synthetic data, augmentation, eval-set generation** — spread and novelty are the goal.

- **Brainstorming genuinely distinct ideas**, or breaking a model out of a repetitive rut.
- **Diverse reasoning paths** for self-consistency voting.

Two rules when you go above 1: (i) **pair it with $\text{top_p} < 1$ or top_k** to clip the long tail that high T fattens; (ii) **never on correctness-bound tasks** — it only buys format breaks and hallucinations.

D The $T \rightarrow \infty$ limit — what “infinity” actually does

The instinct is correct. As $T \rightarrow \infty$ every logit is divided into oblivion, so the exponentials equalise:

$$\lim_{T \rightarrow \infty} p_T(v) = \lim_{T \rightarrow \infty} \frac{e^{L(v)/T}}{\sum_w e^{L(w)/T}} = \frac{e^0}{\sum_w e^0} = \frac{1}{|V|}.$$

The next token is drawn **uniformly at random over the whole vocabulary**, ignoring the prompt entirely — the model’s computation becomes irrelevant. A naïve floating-point implementation gets this right for free: $\text{logit} / \infty = 0$, $\exp(0) = 1$, normalise $\rightarrow$ uniform. That is also why the “crazy” large-but-finite values you can type into llama.cpp already collapse toward uniform: $L(v)/10^9 \approx 0$ for every token.

Three caveats a “correct” implementation must respect:

- **Masking.** Disallowed tokens carry $L = -\infty$; then $-\infty/\infty = \text{NaN}$. A correct kernel pins masked tokens at probability 0, so $T \rightarrow \infty$ is uniform over the *allowed* vocabulary, not literally every id.
- **Truncation samplers run first.** With $\text{top_k} / \text{top_p} / \text{min_p}$ active, the candidate set is still chosen by logit rank, so $T = \infty$ gives uniform over a *context-dependent* shortlist — not fully prompt-independent unless truncation is off.
- **Greedy is untouched.** Argmax decoding is temperature-invariant (Prop 4.1) for *every* finite T , however large — it still returns the model’s top token. Only *stochastic sampling* degenerates to uniform. The two decoders diverge maximally exactly at $T = \infty$.

In the language of §1–§6 this is the bottom row of the table: *uniform mixing*, *maximum entropy*. The aggregate diverges as

$Z_T \approx \max_v L(v) + T \log |V|$, yet the *decision distribution* flattens to $1/|V|$. The winner-versus-field gap that T controls has been opened all the way: every coalition counts equally, so the decode forgets which one was geometrically closest.

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